

Syllabus for MSCF Probability

Summer 2018

Description: This course covers key fundamental concepts of probability, random variables and their distributions, expectations, and conditional probabilities.

Prerequisites: You should be comfortable with calculus, up to and including integration of functions of multiple variables.

Learning Objectives: My objectives for this course are as follows:

1. You will recognize the important role that probability distributions play in the modelling of real (financial) processes.
2. You will learn basic, set-theoretic notions of probability, along with standard ideas of conditional probability, including Bayes' Rule.
3. You will learn the properties of standard "named" distributions.
4. You will have a basic appreciation of some of the more advanced concepts of probability theory that will appear in later courses in the program.
5. You will learn how to work with random variables and their distributions, including how to derive the distribution of a transformation of another random variable or random vector.
6. You will learn how to work with multivariate distributions, including deriving marginal and conditional distributions.
7. You will learn how to work with moment generating functions, and why these functions are useful.
8. You will have a deep understanding of the (univariate and multivariate) normal distribution and its properties.

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Office hours: TBA

Web site: <http://www.cmu.edu/canvas>

TAs: TAs will hold office hours electronically, at a time to be determined.

Calendar: In Pittsburgh, lectures take place in Tepper 145. In New York, lectures are in the Quantbot Classroom.

Lecture 1: July 31, 1:00 to 3:00 - Lecturer in Pittsburgh

Lecture 2: August 1, 1:00 to 3:00 - Pittsburgh

Lecture 3: August 2, 1:00 to 3:00 - Pittsburgh

Lecture 4: August 3, 1:00 to 3:00 - Pittsburgh

Lecture 5: August 7, 1:00 to 3:00 - Pittsburgh

Lecture 6: August 8, 1:00 to 3:00 - Pittsburgh

Lecture 7: August 9, 1:00 to 3:00 - New York

Lecture 8: August 10, **10:30 to 12:30** - New York

Lecture 9: August 14, 1:00 to 3:00 - Problem Session

Lecture 10: August 15, 1:00 to 3:00 - Problem Session

Final Exam: Friday, August 17, noon to 3:00 PM

Course Grade: You are responsible for checking your scores regularly on Canvas, and reporting any errors or discrepancies to me.

Your total score will be based on the four homeworks (30%), the problem sessions (20%) and the final exam (50%). **However, you need to pass the exam (score at least 60%) in order to pass the course.**

Discussion Board: We will utilize the “Discussions” feature of Canvas. This is a means for you to submit questions you have concerning the lecture material, homework assignments, or anything else regarding the course. Click on the “Discussions” button on the left menu to get started.

Homework: There are four homework assignments in this course, due at the **start of lecture** on August 2, August 7, August 9, and August 14.

Handwritten homeworks are fine; of course, they should be clear.

Homework should be submitted by the start of lecture.

The homeworks are a key component of this course, and a large amount of learning will occur during the completion of the exercises. There will be instances where the homework is used to extend material that was taught during the lecture. There will be cases where important results will be presented in the homework. These new results should be considered as being equally important to results presented during lecture.

Further, the difficulty level of the homework exercises is, on average, greater than that of the examples done during lecture. It is not a wise use of our limited lecture time to work through many extended examples, and watching me work an exercise is not a replacement for doing the work yourself.

Problem Sessions: There is no substitute for working through exercises, since one can only learn so much from watching someone else complete an exercise. For this reason, the last two “lectures” will be “problem sessions” during which students will work to complete probability exercises. These exercises will be drawn from past course exams and interview questions, among other sources. Participation in these sessions is required. More details will follow.

Communication: Please do not contact TAs with concerns over grading, or any other issues with the course. Instead, email the instructor.

I assume that you check, at least daily, your official University email (in other words, your `@andrew.cmu.edu` address). Email can be used for important announcements, so make sure that you are regularly reading mail sent this account. Of course, there is a way to forward mail from this account to another account.

You can email me any questions you have regarding the lecture and/or homework. If I feel it is appropriate, I will ask you to post your question to the discussion board. Ultimately, it will be quicker if you post the question directly to the discussion board; I will still see it immediately.

I am good about responding to email, and try to be helpful. **However**, you should be reasonable about expectations of responses to questions close to when homeworks are due.

All communication in this course (verbal and written) between students, instructor, and TAs is required to be done in English.

Final Exam: The final exam will be on Friday, August 17, noon to 3:00 PM, Rooms TBA.

The final exam will be three hours in length. You will be allowed to bring one sheet of **handwritten** notes. (Both sides of an 8.5 by 11 inch sheet of paper.) You will also be provided with a table listing important properties of standard distributions, along with a sheet of useful math facts. Both of these sheets are available on Canvas on the “Files” page.

You will not be allowed to use a calculator or computer on the exam.

If you understand everything that was covered in the lectures, and everything that appeared on the homeworks, then you are ready for the final exam. Review material will be posted on Canvas and included in the Problem Sessions. This material will give you an idea of the **types** of questions that I ask on exams; I will not (intentionally) repeat a question from an old exam.

Lectures: I expect you to attend all lectures. Also, I assume you hear everything I say during lecture. In other words, if I have to make an important announcement, and I do so during normal lecture time, I am not obligated to deliver the announcement via any other means.

The lecture notes will be delivered via the document camera. These will be scanned into a .pdf file and posted on the Canvas site following the lecture. If needed, I will clarify, expand, and/or correct the notes prior to posting them. **You should write the notes yourself during the lecture. In my opinion, there is no substitute for this in the learning process.**

There will be a handout of lecture notes posted on Canvas in advance of the lectures. You should print a copy of this prior to lecture. **These are not the entirety of what we will cover during lecture.**

You are encouraged to ask questions at any time during the lectures. If I feel that your question ventures into technical details which are not pertinent to the discussion, I may ask you to either come to office hours or send me an email. You should not interpret this as me discouraging questions and discussion.

Integrity: Please read CMU's policy on cheating and plagiarism:
<http://www.cmu.edu/policies/documents/Cheating.html>

You are not allowed to utilize any materials from past versions of this course. Any such use will be considered cheating, and treated as such.

Collaboration is a tricky issue. The support and assistance of your classmates can go a long way towards helping you to understand the material. Ultimately, however, you are responsible for preparing yourself for the final exam, and for the later courses that build on probability. I encourage you to collaborate, but, when it comes time to write up your solutions, you should work independently and each student should make sure that he/she understands the results. Instances where students copy the work of another student will be treated as cheating.